

Association Rules Mining on Laptop Market Dataset

Results and Business Insights

1 Introduction

This report presents the results of association rules mining applied to a laptop market dataset from Algeria. The objective was to discover meaningful patterns in laptop specifications, pricing, and market positioning through the application of Apriori and FP-Growth algorithms.

1.1 Dataset Overview

- **Domain:** Laptop market (Algeria)
- **Transactions:** Laptop listings with specifications
- **Features:** CPU tier, GPU tier/family, RAM configuration, storage, display specifications, price ranges, laptop models
- **Preprocessing:** Continuous variables discretized into meaningful buckets; categorical features one-hot encoded

1.2 Experimental Design

Two main experimental configurations were tested:

1. **Experiment 1:** Tier-based approach (simplified features)
 - CPU/GPU classified into performance tiers (High-End, Mid-Range, Entry-Level, Legacy)
 - Focus on general hardware categories
2. **Experiment 2:** Detailed feature approach
 - CPU families (e.g., Intel i5/i7/i9, AMD Ryzen 5/7/9, Apple M-series)
 - GPU families (e.g., NVIDIA RTX 40/50 series, GTX series, Intel Integrated)
 - More granular specification details

Both Apriori and FP-Growth algorithms were applied to each configuration with parameters:

- Minimum support: 5%
- Minimum confidence: 60%

2 Key Findings

2.1 Algorithm Performance

Both Apriori and FP-Growth produced identical rule sets for each experiment, validating the consistency of discovered patterns. Each experiment configuration generated 20 association rules with strong statistical metrics.

2.2 Experiment 1: Best Overall Results

Experiment 1 (tier-based approach) yielded the most diverse and actionable patterns across 5 distinct categories:

Pattern Type	Count	Percentage
CPU-GPU Tier Associations	8	40%
RAM Technology Patterns	5	25%
Storage Configurations	3	15%
Display Specifications	3	15%
Form Factor Patterns	1	5%

Table 1: Distribution of pattern types in Experiment 1

2.3 Statistical Quality Metrics

Metric	Range	Interpretation
Support	5.1% – 9.4%	Substantial market segments
Confidence	62% – 93%	Strong reliability
Lift	3.76 – 8.75	4-9× above random chance
Conviction	2.3 – 13.0	Strong directional relationships

Table 2: Statistical quality of discovered rules (Experiment 1)

All rules demonstrate lift values significantly above 1.0, indicating associations far stronger than random chance. Confidence levels consistently exceed 60%, ensuring reliable pattern predictions.

3 Most Significant Patterns

3.1 Pattern 1: Modern Efficiency Technology (Lift: 8.75)

Rule: {LPDDR5/LPDDR5X RAM} → {Integrated GPU, Mid-Range GPU Tier}

- Support: 5.1% Confidence: 73.9%
- **Insight:** Latest-generation RAM technology strongly correlates with power-efficient integrated graphics
- **Business implication:** Industry trend toward efficiency over raw performance in modern laptops
- **Market segment:** Premium ultrabooks and business laptops

3.2 Pattern 2: Compact Mid-Range Configuration (Lift: 8.09)

Rule: {Mid-Range CPU Tier, 13-14" Screen} → {Mid-Range GPU, Integrated GPU}

- Support: 5.1% Confidence: 68.4%
- **Insight:** Compact form factor laptops with mid-tier CPUs predominantly use integrated graphics
- **Business implication:** Ultrabook/business laptop segment prioritizes portability and battery life
- **Market segment:** Corporate/professional mobile computing

3.3 Pattern 3: MacBook Display Signature (Lift: 9.49)

Rule: {QHD+ Display, 13-14" Screen} → {MacBook}

- Support: 6.9% Confidence: 96.3%
- **Insight:** 96% of laptops with QHD+ resolution in 13-14" form factor are MacBooks
- **Business implication:** Apple dominates the premium compact laptop display segment
- **Competitive gap:** Limited non-Apple alternatives with equivalent display quality

3.4 Pattern 4: Gaming Laptop Standard (Lift: 5.29)

Rule: {257-512GB SSD, Dedicated GPU} → {Mid-Range GPU, FHD Display}

- Support: 6.4% Confidence: 62.1%
- **Insight:** Entry-level gaming laptops typically feature FHD displays (not 4K) and mid-range dedicated GPUs
- **Business implication:** Gaming market prioritizes frame rate over resolution; 256-512GB storage is standard
- **Market segment:** Entry-to-mid gaming laptops (e.g., RTX 3060/4060 class)

3.5 Pattern 5: Gaming RAM Standard (Lift: 4.15)

Rule: {12-18GB RAM, Mid-Range GPU, FHD} → {Dedicated GPU}

- Support: 6.9% Confidence: 82.0%
- **Insight:** 16GB RAM is the de facto standard for laptops with dedicated graphics
- **Business implication:** Below 8GB is insufficient; above 32GB is overkill for most gaming use cases
- **Market segment:** Gaming and content creation laptops

4 Market Segmentation Insights

The discovered patterns reveal three distinct market segments:

1. Premium Ultraportable (Apple-dominated)

- Configuration: 13-14", QHD+, integrated GPU, LPDDR5, mid-range CPU
- Characteristic: MacBook near-monopoly (96.3% confidence)
- Opportunity: Market gap for premium non-Apple alternatives

2. Mid-Range Gaming/Creator

- Configuration: 14.1-15.6", FHD (144Hz), dedicated GPU, 16GB RAM, 256-512GB SSD
- Characteristic: Balanced performance for gaming and content creation
- Opportunity: Most competitive segment; standardization benefits inventory

3. Business/Productivity Ultrabook

- Configuration: 13-14", FHD, integrated GPU, LPDDR5, mid-range CPU
- Characteristic: Efficiency-focused, MacBook Air competitor
- Opportunity: Windows alternatives emphasizing battery life and portability

5 Experiment 2 Results: Apple Product Consistency

Experiment 2 (detailed CPU/GPU families) exclusively discovered Apple/MacBook-related patterns. This finding reveals:

- **Apple’s Strategy:** Highly consistent product configurations enable pattern discovery at granular levels
- **Competitor Challenge:** High SKU variability in non-Apple brands prevents detailed pattern emergence
- **Business Insight:** Reduced product complexity strengthens brand identity and market positioning

6 Actionable Recommendations

6.1 For Retailers

1. **Inventory Management:** Prioritize standard configurations (16GB RAM with dedicated GPU; 256-512GB SSD)
2. **Product Bundling:** Create segment-specific packages
 - Gaming: 15.6” + FHD + mid-range GPU + 16GB
 - Business: 13-14” + LPDDR5 + integrated GPU
 - Premium: 13-14” + QHD+ (MacBook alternative positioning)
3. **Pricing Strategy:** Align pricing tiers with discovered segments (efficiency/gaming/premium)

6.2 For Manufacturers

1. **SKU Optimization:** Reduce product variation; standardize proven configurations
2. **Competitive Positioning:** Address premium compact display gap (compete with MacBook on QHD+ quality)
3. **Technology Adoption:** Accelerate LPDDR5/DDR5 adoption as market differentiator
4. **Gaming Focus:** FHD 144Hz as standard (not 4K); mid-range GPU sweet spot

7 Methodology Insights

7.1 Data Quality Impact

Initial experiments included features with "Unknown" values, resulting in 40% meaningful rules. After removing these columns, rule quality improved to 100% meaningful patterns. This demonstrates:

- Critical importance of data cleaning in association rule mining
- Missing value handling significantly affects pattern quality
- Trade-off between feature completeness and pattern interpretability

7.2 Feature Granularity Balance

- **Optimal granularity:** Tier-based approach (Experiment 1) achieved best diversity
- **Over-specification:** Detailed features (Experiment 2) captured only consistent Apple patterns
- **Lesson:** Feature engineering requires balancing detail with pattern generalizability

8 Conclusions

Association rules mining on the laptop market dataset successfully identified:

1. **Three distinct market segments** with characteristic configurations
2. **Industry trends:** Shift toward power efficiency (LPDDR5 + integrated GPU)
3. **Competitive landscape:** Apple’s premium compact display dominance (96% market share in segment)
4. **Standard configurations:** Gaming (16GB + FHD + mid-GPU), Business (13-14” + LPDDR5 + integrated)
5. **Strategic gaps:** Limited non-Apple premium ultraportable options

All discovered rules demonstrate strong statistical significance (lift 3.5-9.5) and practical business value. The tier-based approach (Experiment 1) proved most effective for discovering diverse, actionable patterns suitable for inventory planning, product bundling, and competitive strategy.

8.1 Key Takeaway

The analysis reveals that **feature engineering granularity critically impacts pattern discovery quality**. Appropriately aggregated features (tiers) enable diverse pattern discovery across product categories, while overly detailed features only capture patterns in highly standardized product lines (e.g., Apple).